

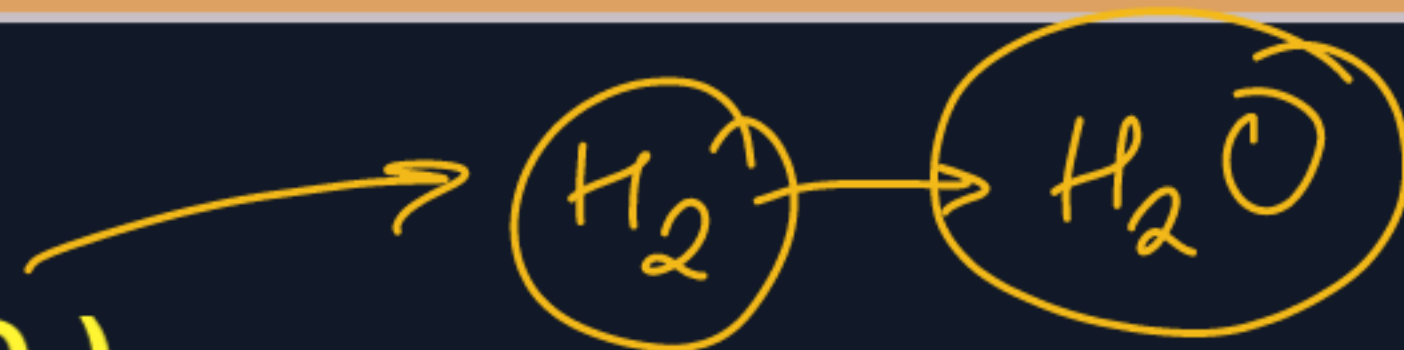
L3

CLASS X - SCIENCE

METALS AND NON-METALS

PRASHANT KIRAD

Nitric Acid (HNO_3)



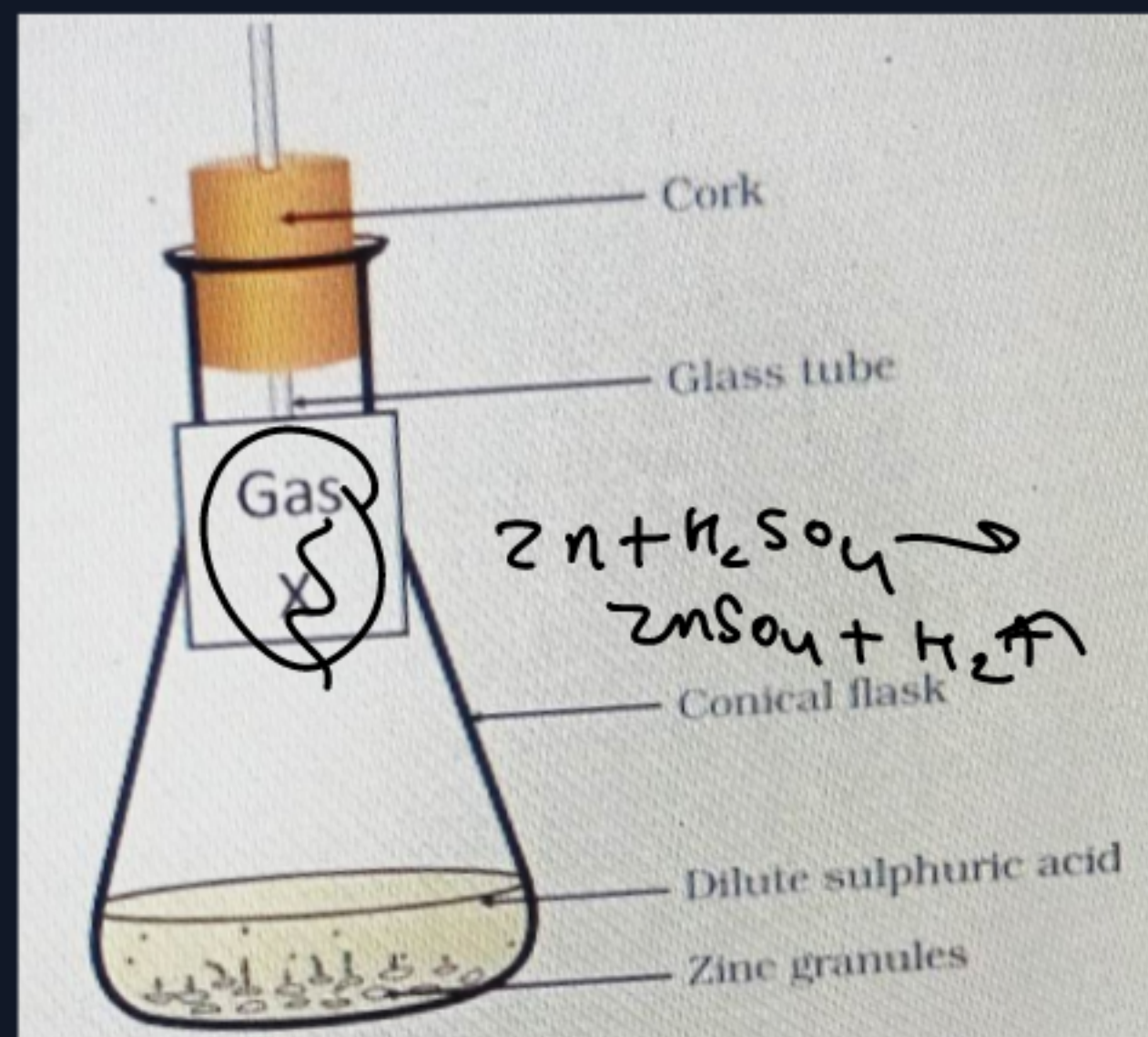
- Hydrogen gas (H_2) is not evolved when a metal reacts with nitric acid (HNO_3). This is because HNO_3 is a strong oxidizing agent. It oxidizes the H_2 produced to water and itself gets reduced to any of the nitrogen oxides (N_2O , NO , NO_2).
- However, magnesium (Mg) and manganese (Mn) react with very dilute HNO_3 to evolve H_2 gas.



Check yourself

Q. Which of the following tests is given by the gas 'X' produced in the set up shown?

- ☒ a) Limewater turns milky white
- ☐ b) Puts off a burning matchstick
- ☒ c) Burning matchstick produces a pop sound and the flame goes out.
- ☐ d) Choking smell with the smell of burning Sulphur

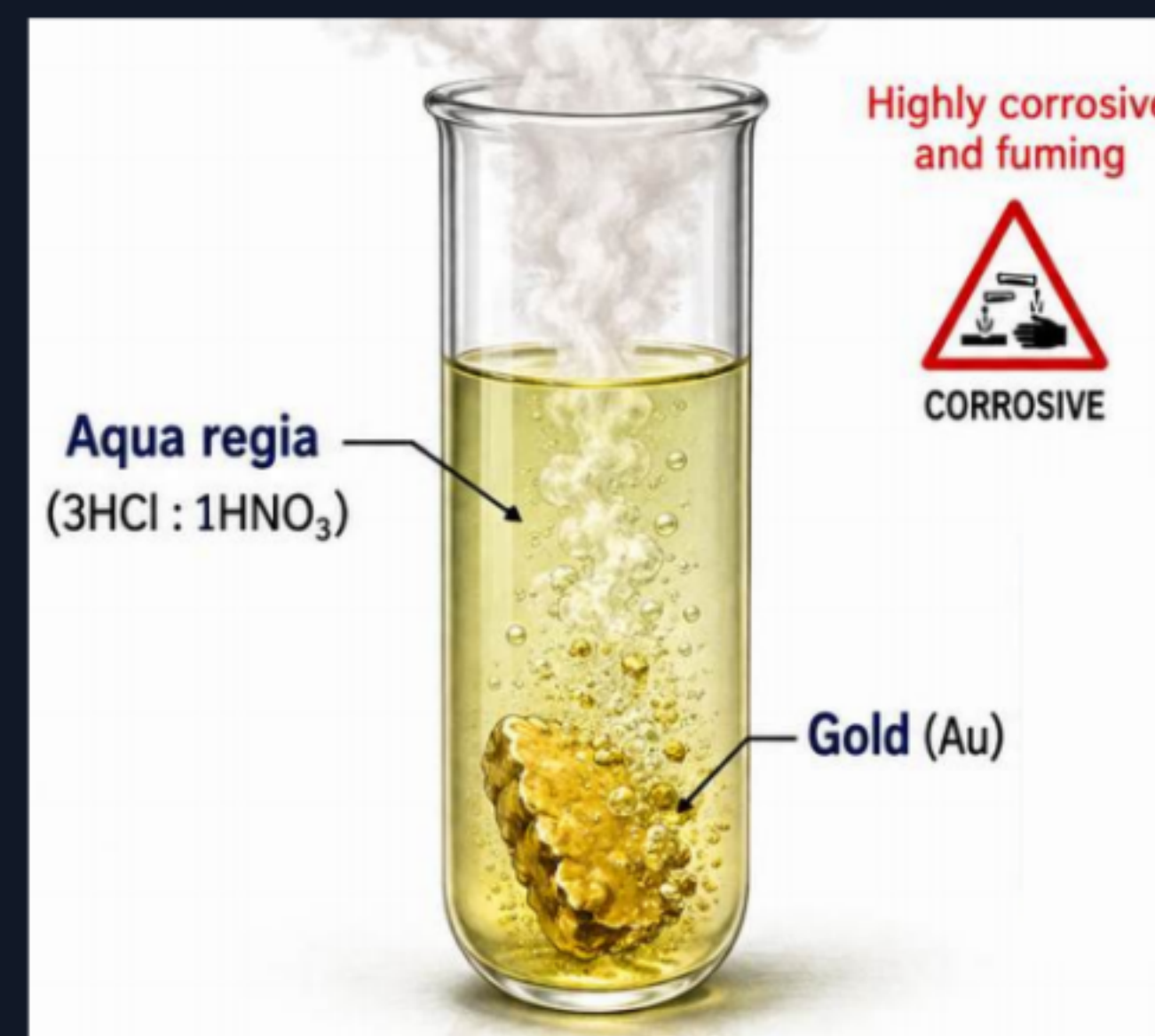


AQUAREGIA

- Aqua regia is a freshly prepared mixture of concentrated hydrochloric acid (HCl) and concentrated nitric acid (HNO₃) in the ratio of 3 : 1.

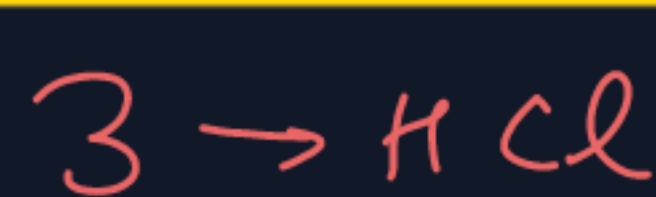
Concentrated Hydrochloric acid (HCl) : Concentrated Nitric acid (HNO₃) = 3 : 1

- It is a highly corrosive, fuming liquid capable of dissolving even noble metals like gold and platinum which do not dissolve in either acid alone.
- In this process, HCl provides chloride ions and HNO₃ acts as an oxidising agent. Nitric acid oxidises metals and gets itself reduced to oxides of nitrogen such as NO, NO₂, N₂O.
- Application. Used in purification and testing of gold and platinum in laboratories and jewellery shops.



Check yourself

Q.(a) What is aqua regia? Write its composition.



(b) Name one metal which can be dissolved in aqua regia but not in any single acid.

(c) Write one use of aqua regia.



Check yourself

QUES: A student accidentally mixes concentrated HCl and Concentration HNO_3 in the ratio 1:3.

which statement is most appropriate?

- a) It behaves exactly like aqua regia.
- ☒ b) It cannot dissolve gold as efficiently because the required composition is disturbed.
- c) It becomes a neutral solution
- d) It behaves like a dilute nitric acid.



Check yourself

QUES: A student performs the following experiment:

- Test tube A contains concentration HCl.
- Test tube B contains concentration HNO_3
- Test tube C contains freshly prepared aqua regia.

A small gold strip is immersed in each tube for 39 minutes. Which observation is most likely?

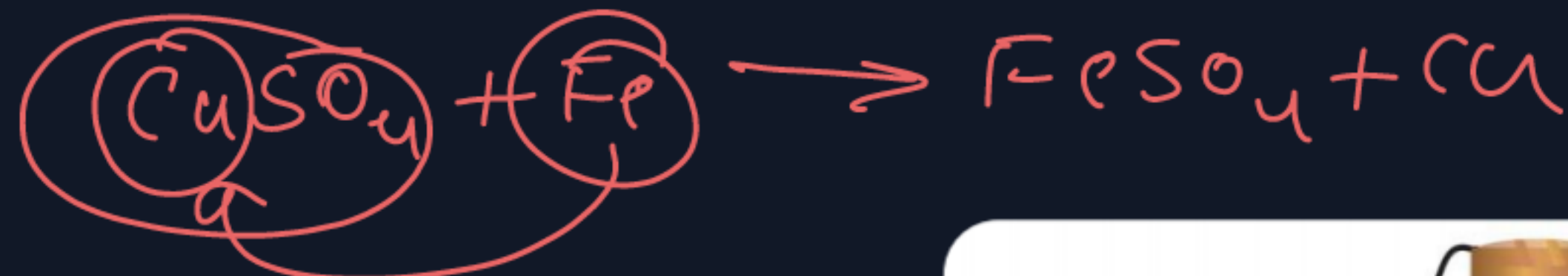
- A) Gold dissolves only in A
- B) Gold dissolves only in C
- c) Gold dissolves equally in all three.
- d) Gold dissolves only A and B but not C.



REACTION OF METALS WITH OTHER SALT SOLUTION



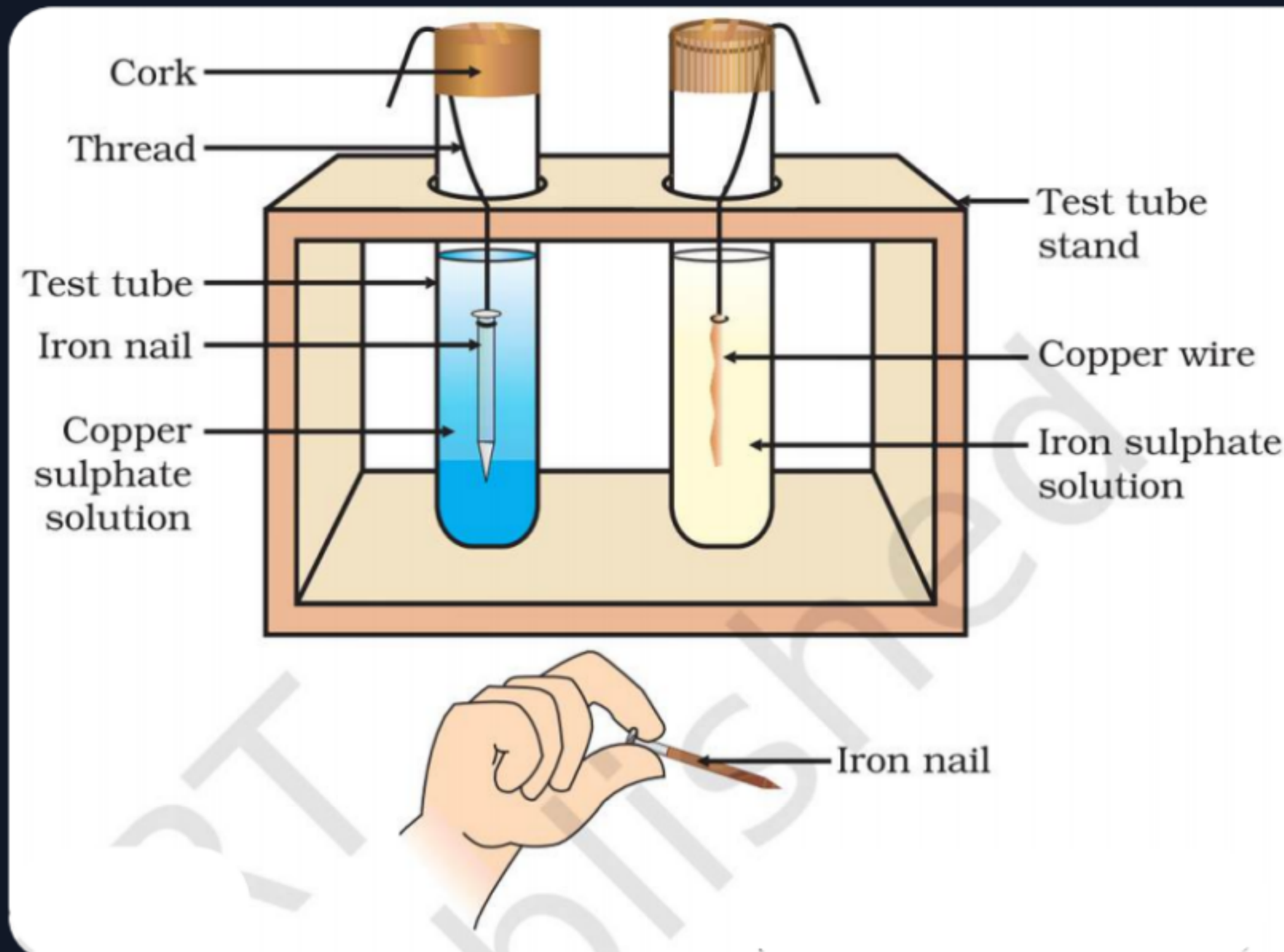
A more reactive metal displaces a less reactive metal from its salt solution.



- Copper wire in $\text{FeSO}_4 \rightarrow$ No reaction (copper less reactive).
- Iron nail in $\text{CuSO}_4 \rightarrow$ Brown deposit of copper on iron nail + solution colour changes (blue fades).



Iron displaces copper because Fe is more reactive than Cu.



Check yourself

QUES: Which one of the following will not occur because the metal involved is less reactive than the metal present in the salt solution?

- a) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
- b) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$
- c) $\text{Cu} + \text{ZnSO}_4 \rightarrow \text{no reaction}$
- d) $\text{Mg} + \text{CuSO}_4 \rightarrow \text{MgSO}_4 + \text{Cu}$



Check yourself

QUES: Which of the following combinations will undergo a displacement reaction?

- ~~(a)~~ NaCl solution + Cu
- ~~(b)~~ MgCl₂ solution + Al
- ~~(c)~~ FeSO₄ solution + Ag
- (d) AgNO₃ solution + Cu



Check yourself

QUES: A metal M displaces iron from an aqueous solution of ferrous sulphate but fails to do so from an aqueous solution of aluminium sulphate.

The metal M is:

- (a) Magnesium
- (b) Copper
- (c) Lead
- (d) Zinc

K
Na
Ca
Mg
Al
✓ Zn
Fe



Check yourself

QUES. An iron strip is immersed in a blue copper sulphate solution. After a few minutes, the blue colour of the solution gradually changes to green, and a reddish brown coating appears on the iron strip. Which statement best explains this observation?

- A. Iron is less reactive than copper
- B. Copper is displaced by iron
- C. Iron reacts with water present in solution
- D. Iron dissolves without undergoing any chemical reaction



Check yourself

Assertion (A): Iron can displace copper from copper sulphate solution.

Reason (R): Iron is more reactive than copper.

- a) Both A and R are true, R is the correct explanation of A.
- b) Both A and R are true, R is the not correct explanation of A.
- c) A is true but R is false.
- d) R is true but A is false.



Check yourself

Ques. A metal R displaces metal Q from the compound PQ:

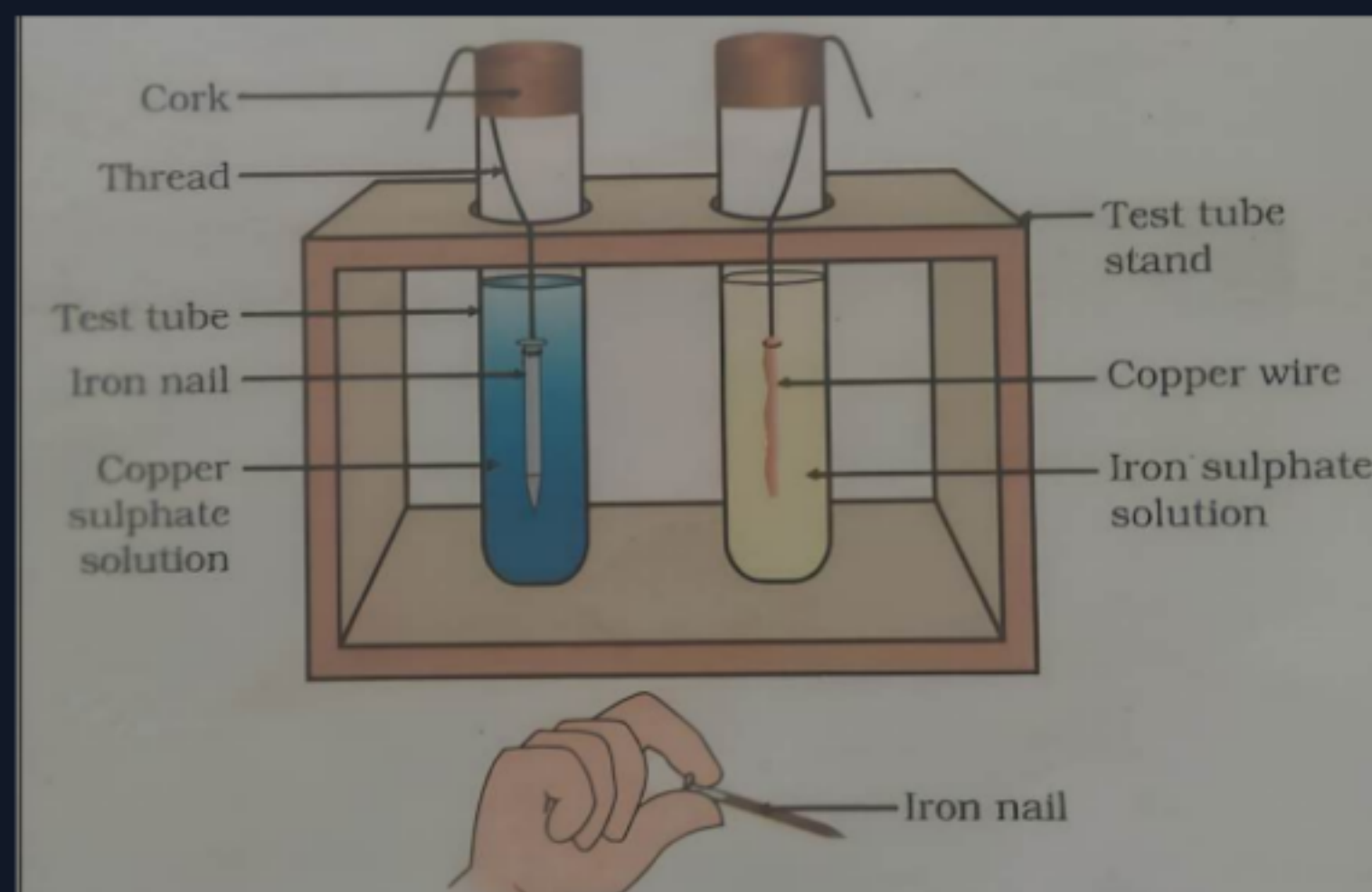


- (a) Which is more reactive, R or Q?
- (b) What type of ion is R likely to form if R is a metal?
- (c) What type of reaction is this?



Check yourself

QUES. Observe the two test tubes A and B in the diagram given below and answer the following questions:



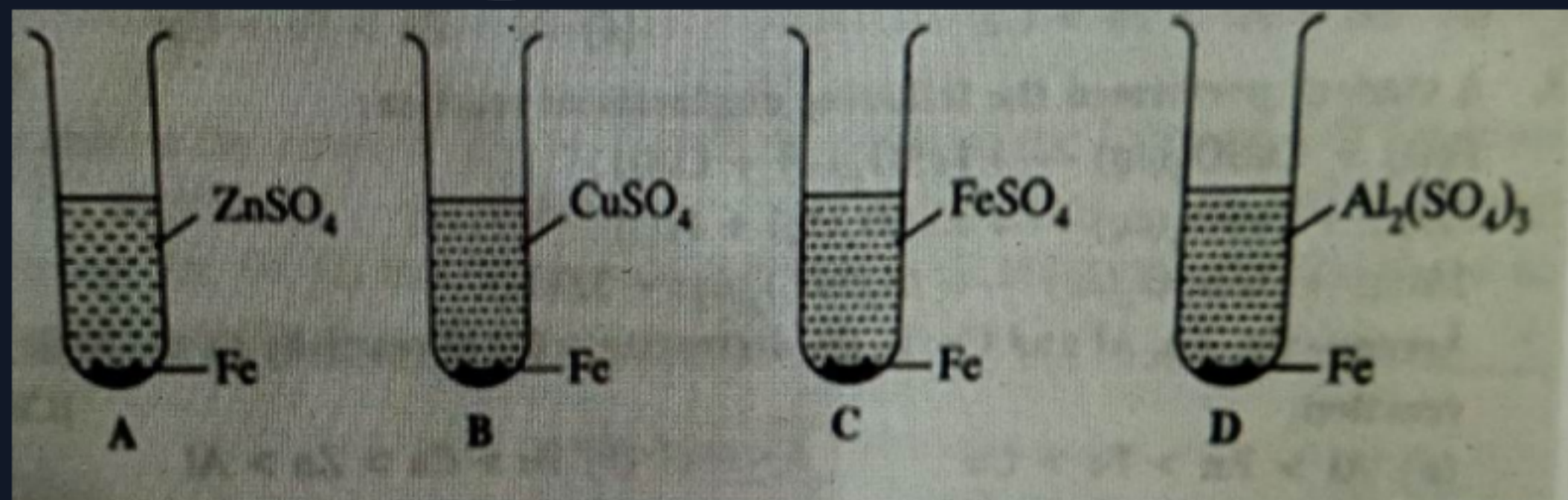
- In which test tube will the reaction take place?
- Write a balanced equation for the reaction?
- Name the type of reaction.



Check yourself

Q. Sakshi was comparing the reactivity of different metals for her science project. She added iron filings in four test tubes, A, B, C, and D, containing aqueous solutions of ZnSO_4 , CuSO_4 , FeSO_4 and $\text{Al}_2(\text{SO}_4)_3$, respectively, as shown in the figure.

(B)



- In which test tube will a displacement reaction occur? Give a reason and write the balanced chemical equation."
- What is the reason for her observation?
- Write a well-balanced equation of the reaction in (b)



Check yourself

Q. Four metals A, B, C and D are added separately to solutions of iron(II) sulphate, copper(II) sulphate, zinc sulphate and silver nitrate. The observations are:

$D \rightarrow C \rightarrow A \rightarrow B$

Metal	FeSO ₄	CuSO ₄	ZnSO ₄	AgNO ₃
A	No reaction	Displacement	X —	—
B	Displacement	✓	No reaction	✓ —
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

- Which of A, B, C and D is the most reactive metal?
- What would you observe if metal B is added to CuSO₄ solution?
- Arrange A, B, C and D in decreasing order of reactivity.

$B > A > C > D$





CHEMICAL PROPERTIES OF NON METALS:

Chemical Properties of **NON METALS**:

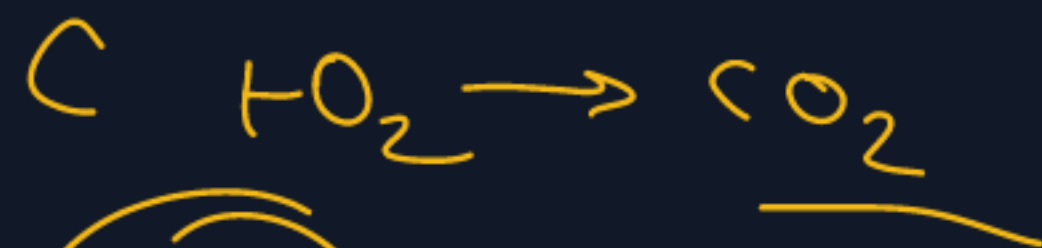
1. Reaction with Oxygen

→ ACIDIC in nature

Non-metal + Oxygen → Non-metal oxide

Example:

- $\text{C (s)} + \text{O}_2 \text{ (g)} \rightarrow \text{CO}_2 \text{ (g)}$
- $\text{S (s)} + \text{O}_2 \text{ (g)} \rightarrow \text{SO}_2 \text{ (g)}$



- *Non-metallic oxides are acidic in nature: CO_2 , SO_2 , SO_3 , etc.*
- *Non-metallic oxides can also be neutral in nature: CO , H_2O , N_2O , etc.*

2. Reaction of non-metal oxide with water

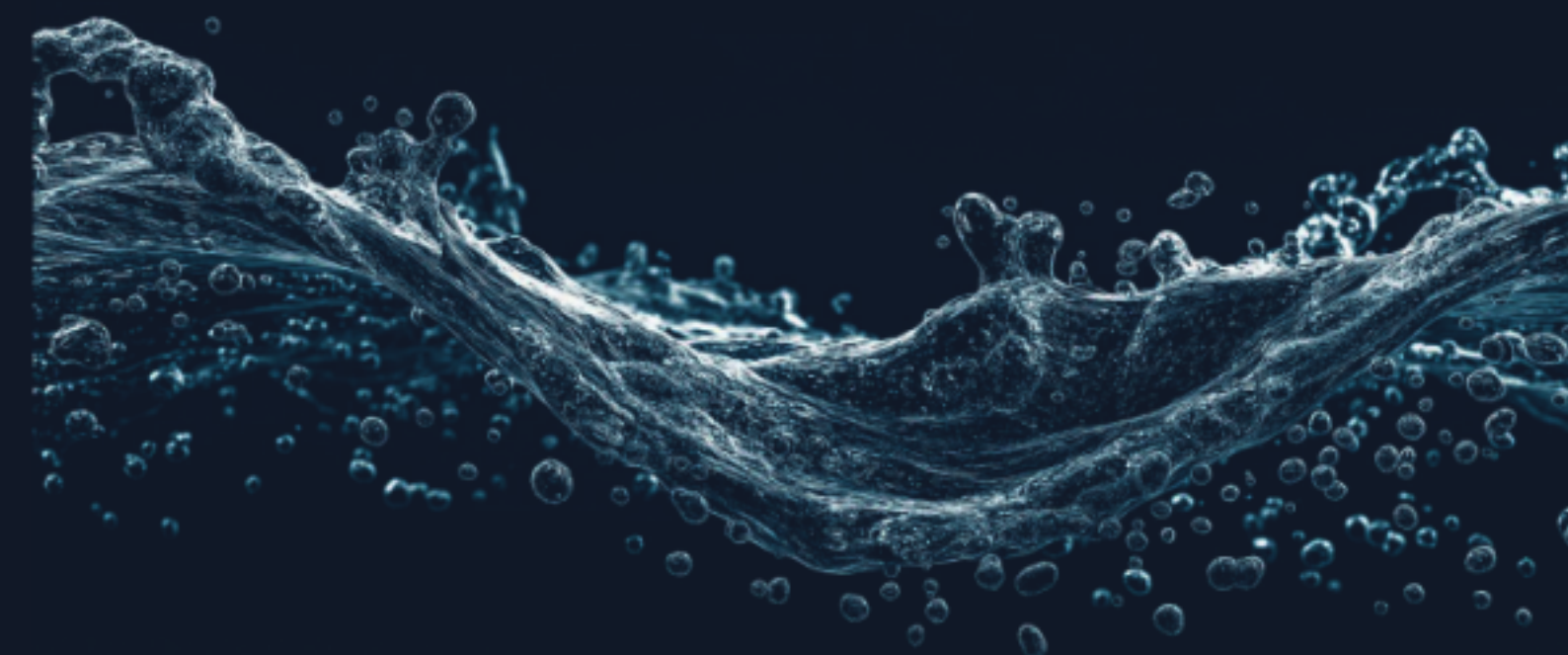
Generally non-metals do not react with water because they cannot displace hydrogen from water



Non-metal oxide + Water → Acid (forms acidic solution)

Example:

- $\text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{H}_2\text{CO}_3 (\text{aq})$
- $\text{SO}_3 (\text{g}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{H}_2\text{SO}_4 (\text{aq})$
- $2\text{NO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{HNO}_2 (\text{aq}) + \text{HNO}_3 (\text{aq})$



Check yourself

Ques. Which of the following is an acidic oxide?

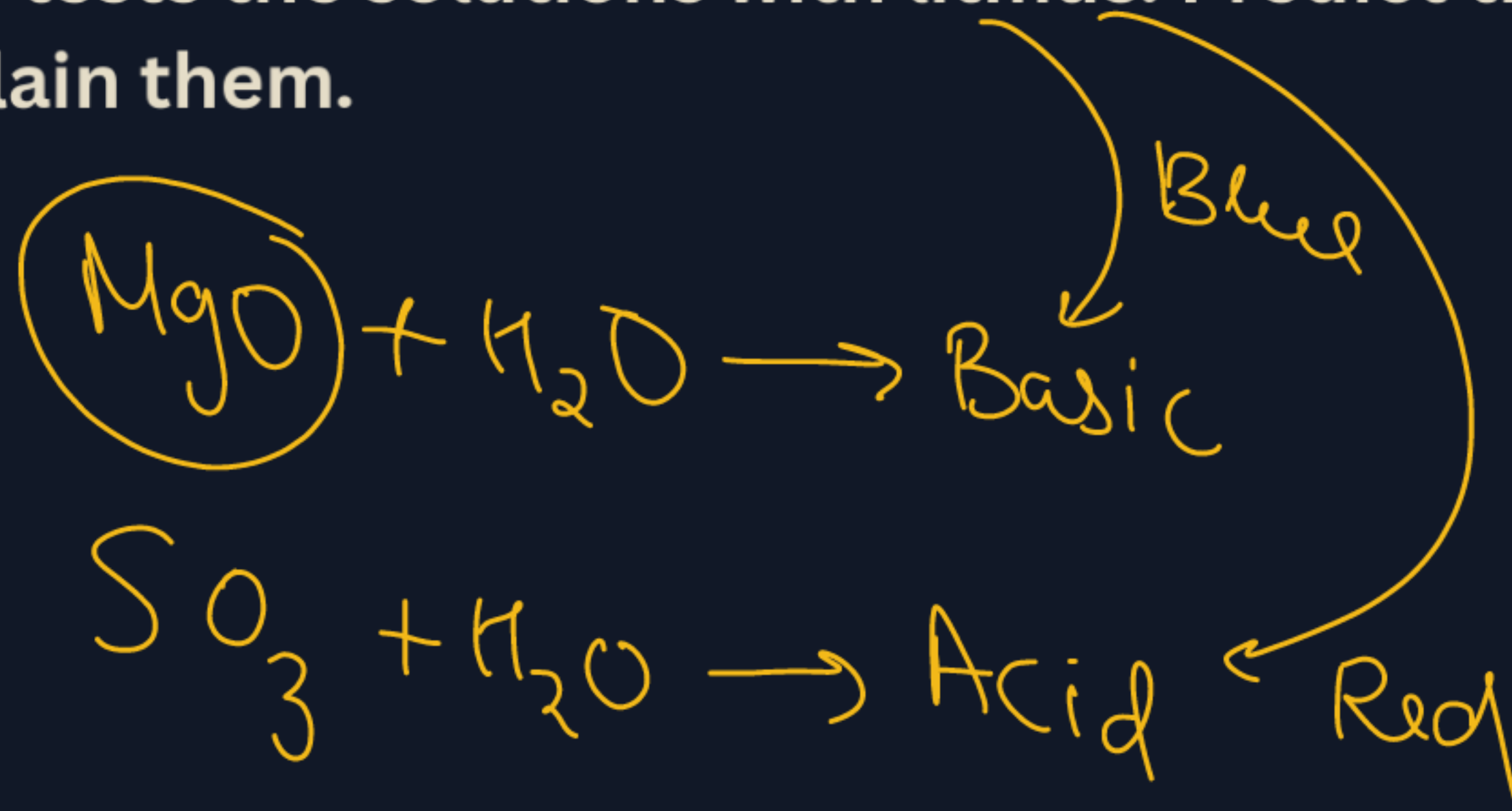
- (a) MgO
- (b) Na_2O
- ☒ (c) SO_2
- (d) CaO

Non-Metal
oxide



Check yourself

Ques. A student burns magnesium and sulphur separately, dissolves their products in water and tests the solutions with litmus. Predict the observations and explain them.





REACTION OF METALS WITH NON METALS

Reaction of Metals with Non Metals

Atoms are most stable when their outermost shell is completely filled.

To become stable:

- Metals lose electrons → form cations.
- Non metals gain electron → form anions.

This transfer of electrons lead to the formations of **ionic compounds**.

② **Ionic compounds:** Compounds formed by the transfer of electrons from a metal to a non-metal, resulting in the formation of positively charged ions (cations) and negatively charged ions (anions) held together by strong electrostatic forces of attraction.



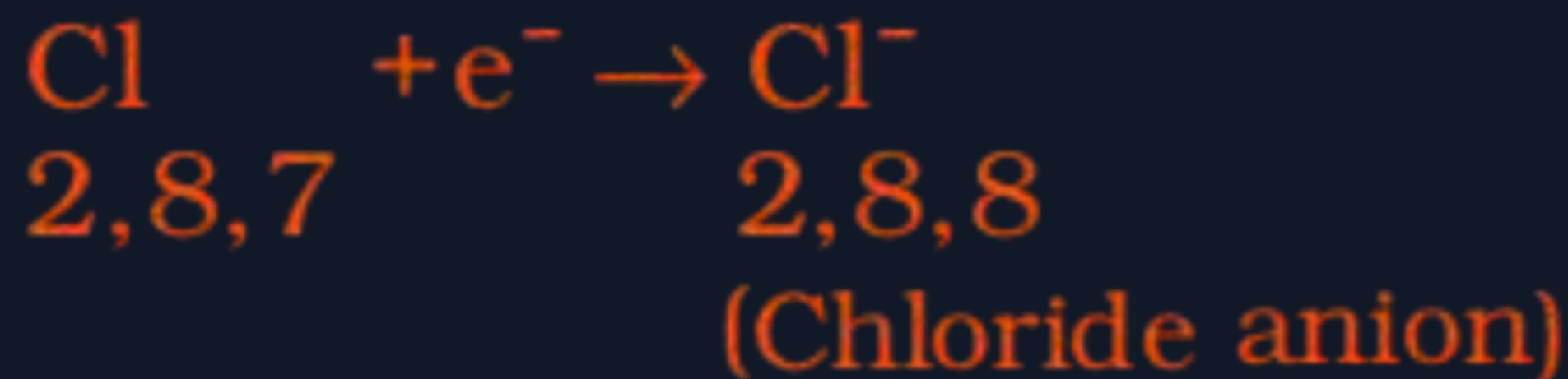
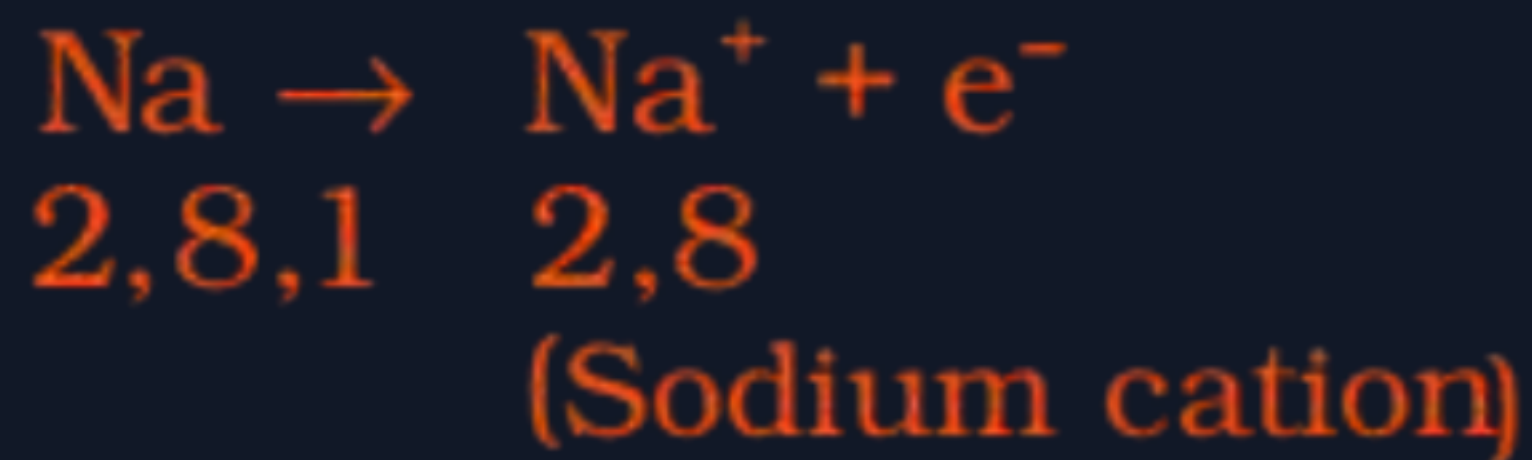


Type of element	Element	Atomic number	Number of electrons in shells			
			K	L	M	N
Noble gases	Helium (He)	2	2			
	Neon (Ne)	10	2	8		
	Argon (Ar)	18	2	8	8	
Metals	Sodium (Na)	11	2	8	1	
	Magnesium (Mg)	12	2	8	2	
	Aluminium (Al)	13	2	8	3	
	Potassium (K)	19	2	8	8	1
	Calcium (Ca)	20	2	8	8	2
Non-metals	Nitrogen (N)	7	2	5		
	Oxygen (O)	8	2	6		
	Fluorine (F)	9	2	7		
	Phosphorus (P)	15	2	8	5	
	Sulphur (S)	16	2	8	6	
	Chlorine (Cl)	17	2	8	7	

Electronic configurations of some elements

The arrangement of electrons in various shells (K, L, M, N, ...) of an atom is called its electronic configuration.

Formation of Sodium Chloride (NaCl)

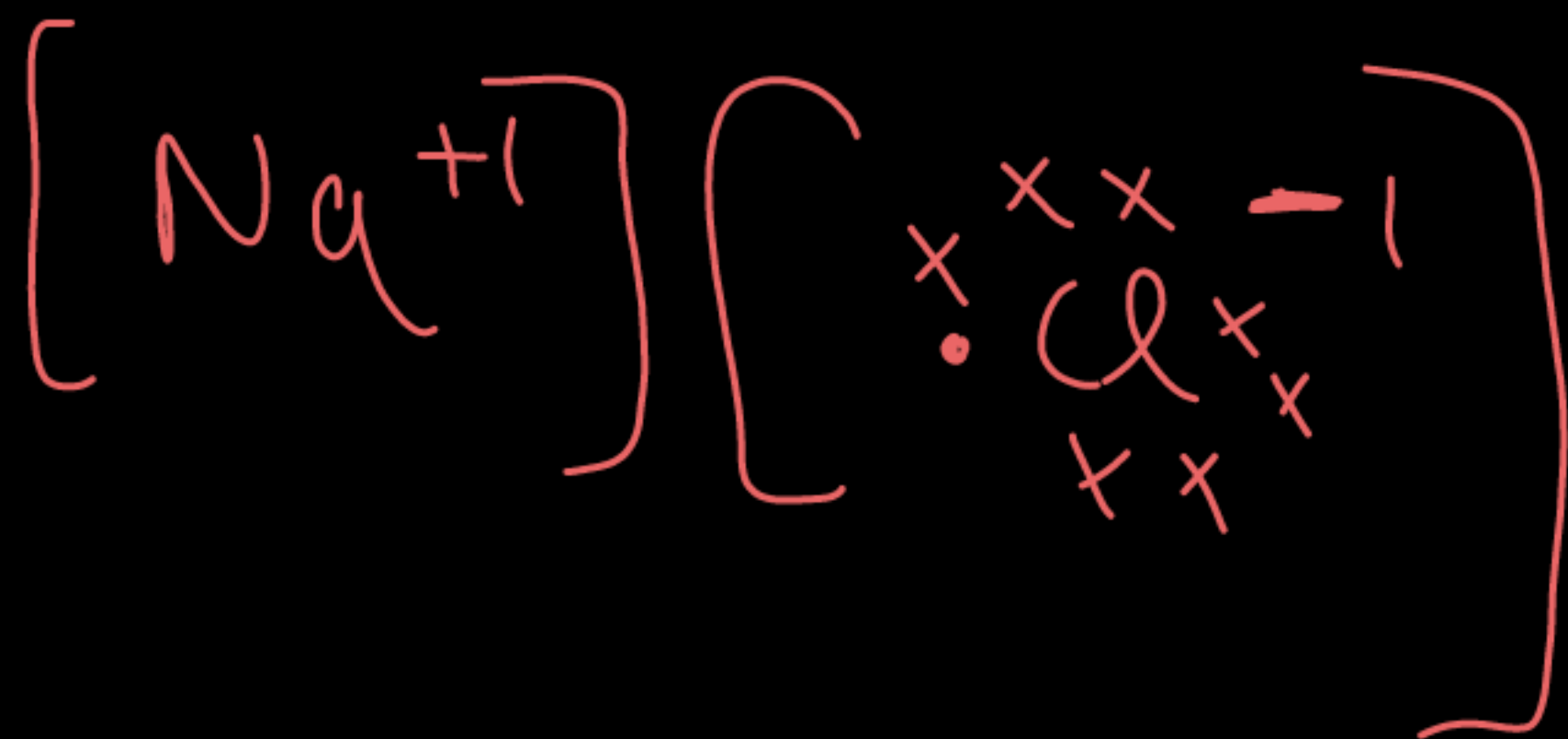
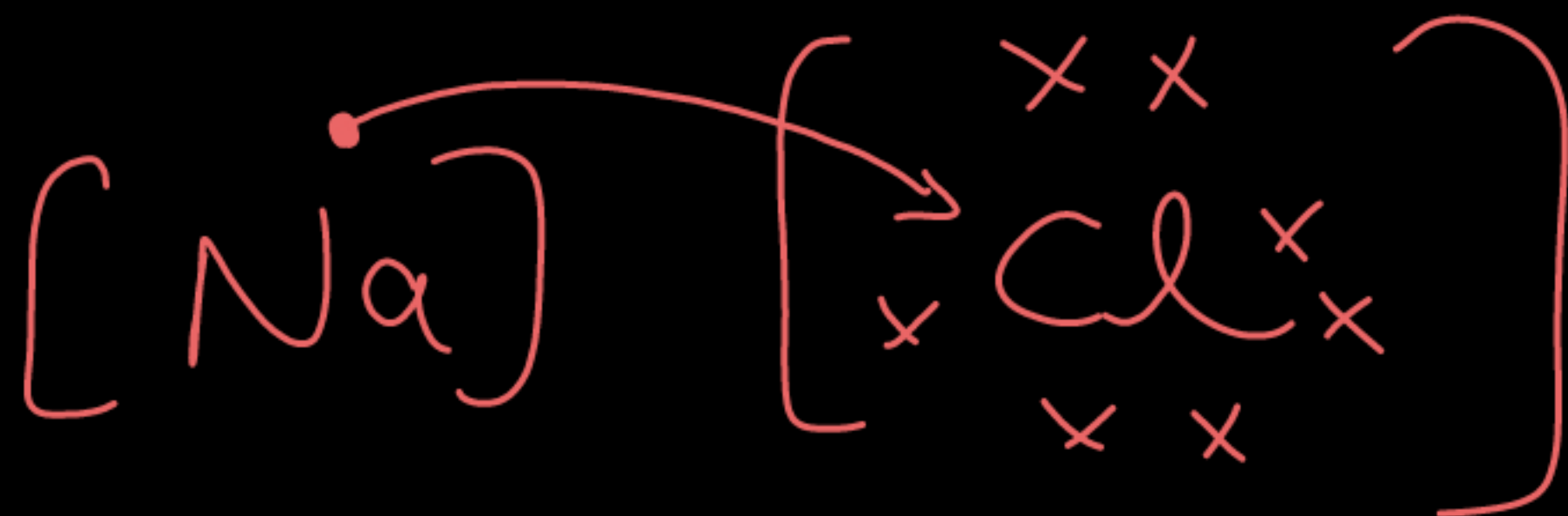


QUES: Explain the formation of sodium chloride with suitable electron transfer diagram.
[NCERT]

Lewis dot Structure

Electron Dot Structures
or Lewis Dot Structures

Nacl



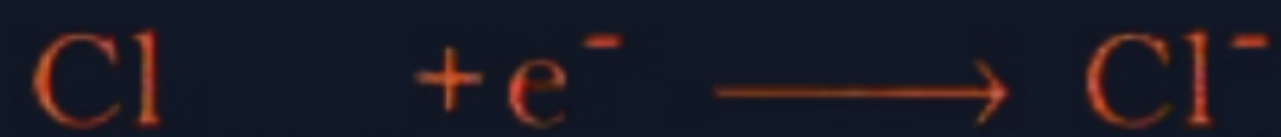
Formation of Magnesium Chloride (MgCl_2)



2, 8, 2

2, 8

(Magnesium cation)

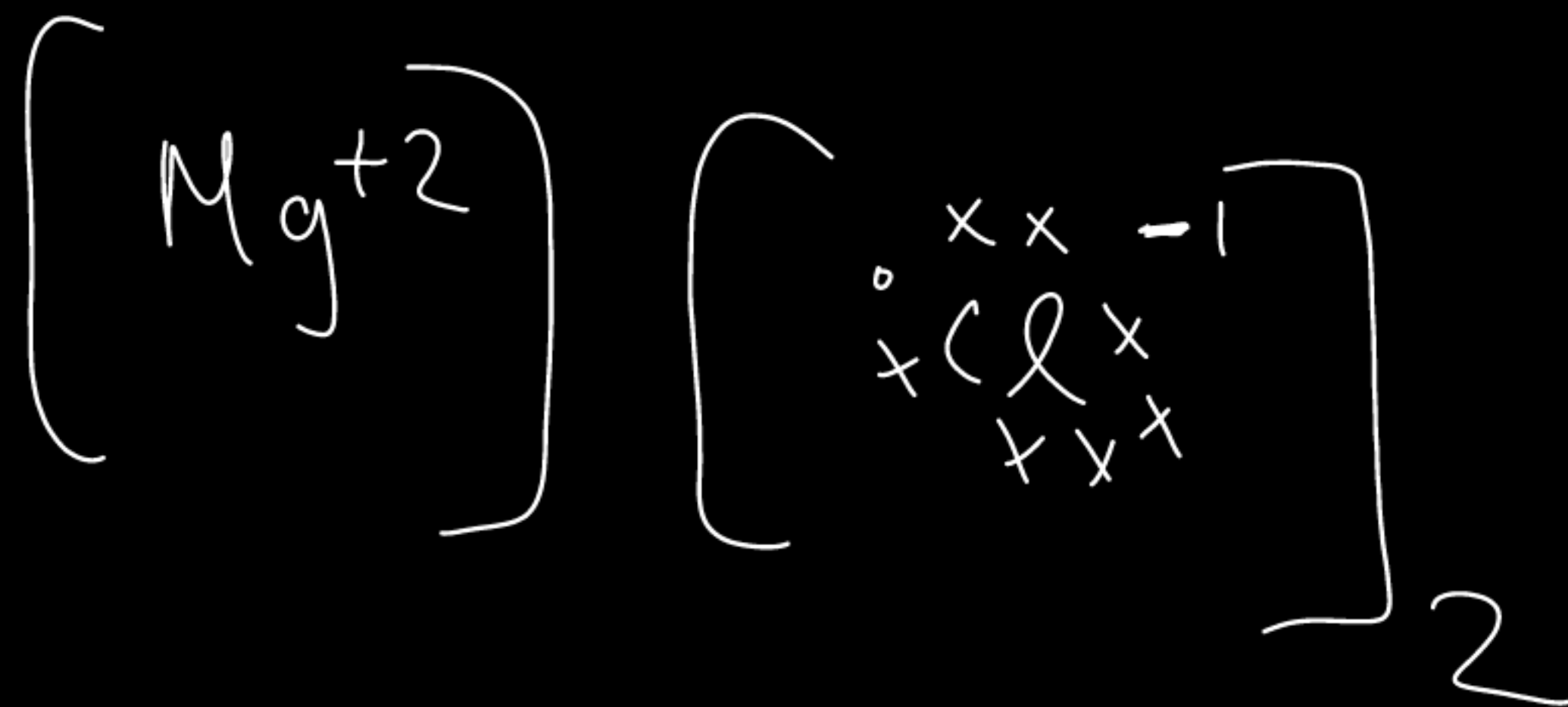
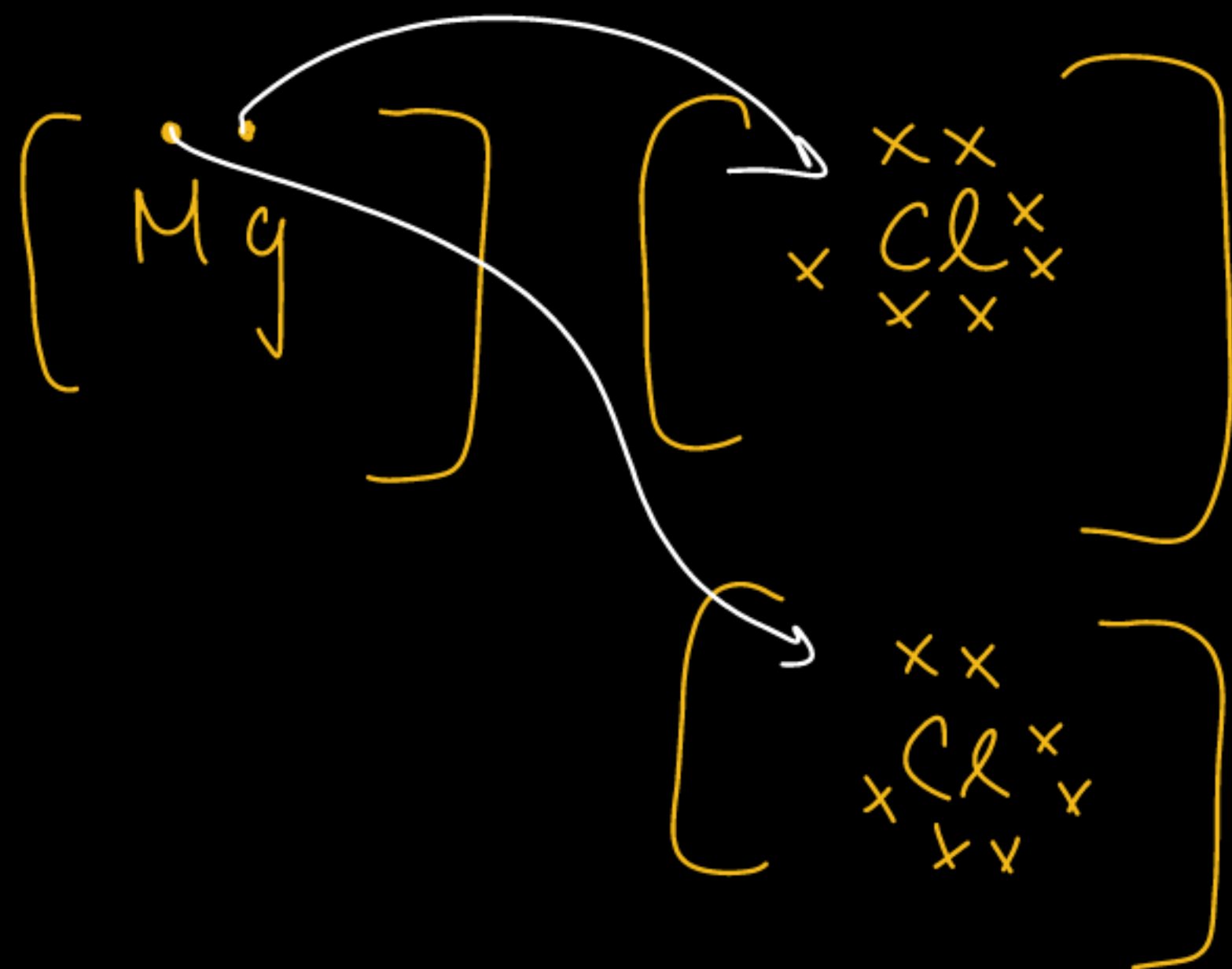


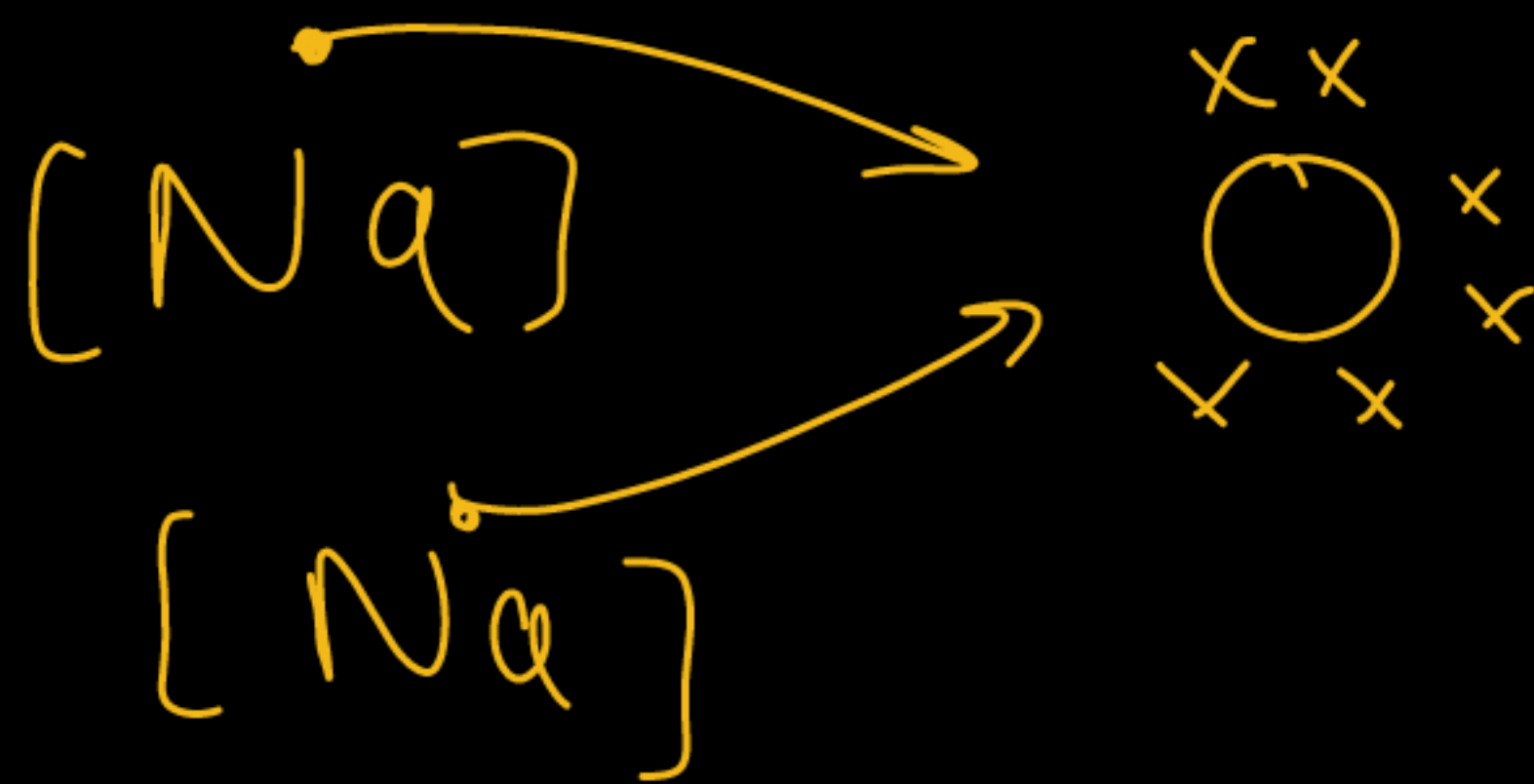
2, 8, 7

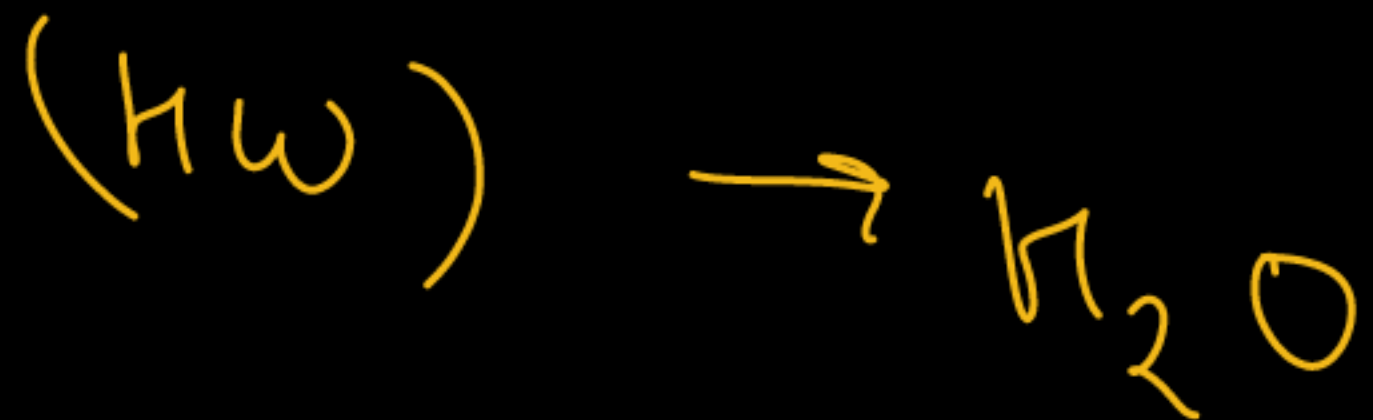
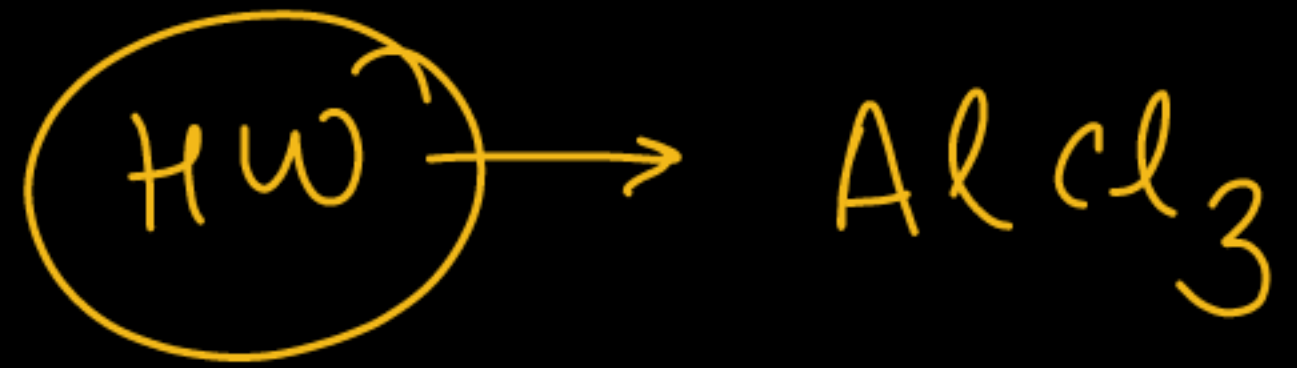
2, 8, 8

(Chloride anion)









Check yourself

QUES. Sodium has electronic configuration 2,8,1 while chlorine has 2,8,7.

Answer the following:

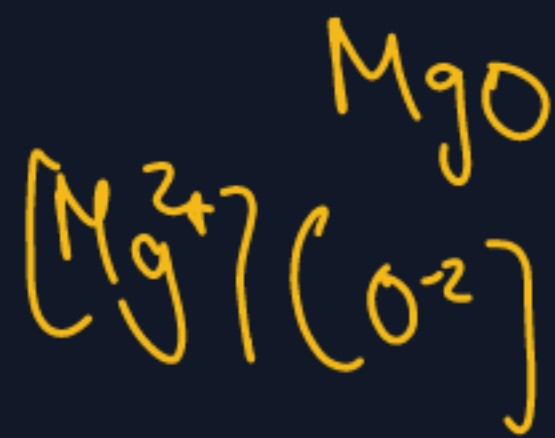
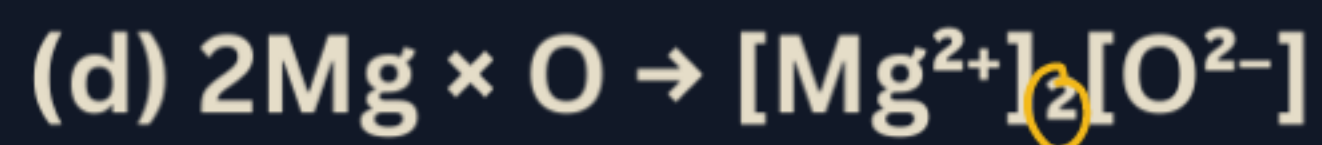
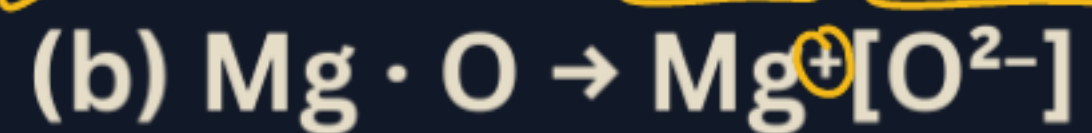
- a. Which item loses electrons? → Na
- b. Which item gains electrons? → Cl
- c. Which ions are formed? →
- d. Name the type of bond present.

Ionic Bond



Check yourself

QUES. The formation of magnesium oxide is correctly shown in option:



Check yourself

QUES. Sodium atom loses one electron during the formation of ionic compound. Explain why it becomes positively charged after losing this electron.



Check yourself

QUES: During the formation of ionic compounds, atoms form either cations or anions. Explain the difference between these two types of ions with one suitable example each.



Check yourself

QUES: (i) Show the formation of Na_2O and MgO by transfer of electrons.
(ii) What ions are present in these compounds?





Milte h Kal!